GRIFFIN LOWELL

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EDUCATION

University of Wisconsin–Madison — Madison, WI **Expected May 2028**

Bachelor of Science, Mechanical Engineering | Certificate in Computer Sciences
GPA: 3.41/4.00 | Dean's Honor List (2 terms)

EXPERIENCE

Engineering Intern — The Technology House, Streetsboro, OH **May 2026 – Aug 2026**

- Operated Carbon and HP 3D printers daily, gaining firsthand experience with resin handling, cleaning, finishing, and quality control
- Identified an 18 minute cassette cleaning routine and a resin leak path through daily printer operation and proposed an out of machine height adjustable, tiltable, and rotatable fixture
- Modeled and manufactured locating hardware that directly mates to the underside of the cassette so it's compatible with hand off cart and locked in place during rotation
- Selected a motorized base with the engineering team to simplify the design and eliminate manual lifting
- Assembled and tested the fixture, presented the results to the engineering team, explaining an existing pour containment issue, and proposed a clip on spout solution

Head of Sailing — Camp Kawaga, Minocqua, WI **June 2023 – Aug 2025**

- Managed the sailing program for three summers, coordinating daily instruction and free sailing, overseeing on water safety, and organizing an inter camp regatta
- Taught beginner sailors boat handling and on water safety through daily instruction

EXTRACURRICULARS

Rotary Inverted Pendulum — Faculty-supervised controls project, UW–Madison **May 2026 – Present**

- Brought a WHEELTEC rotary pendulum provided by a UW–Madison professor to balance using vendor firmware after diagnosing a 5 V adapter mismatch and switching to 12 V
- Reduced oscillation and shakiness by lowering the loop time, adding a low-pass filter, and recalculating the correct upright position
- Added the kick-up feature the vendor only designed for a different pendulum by changing the control handoff from PID to LQR and the exact angle the handoff occurred

PROJECTS

Spread Trader Bot — Python, Charles Schwab API **May 2026 – Present**

- Built and operate a live Python trading system that translates a partner's options strategy into automated orders through the Charles Schwab API
- When live fills lagged the dry run, retuned midpoint pricing and added a retry loop, raising daily fill volume from roughly 30 trades to over 140 and peak daily collected credit to $1,460
- Automated trade logging to record every fill and its collected credit

Raspberry Pi Door Buzzer — SolidWorks, 3D printing, Python **Oct 2025 – Dec 2025**

- Designed a 3D-printed rack and pinion actuator for an apartment entry buzzer with Wi-Fi remote unlock through a Raspberry Pi server

SKILLS

Software: SolidWorks (CSWA Certified), FEA, MATLAB, Python, Java, Git, Excel
Shop and Hardware: Carbon DLS, HP Multi Jet Fusion, FDM 3D printing, laser cutting, mill, lathe, CNC lathe, bead blasting, vapor smoothing, Raspberry Pi, soldering
Interests: Marathon running (3:28 Madison Marathon), weightlifting, nutrition, DJing